From Joint-Embedding Prediction to World Models: A Survey

Ziying Song*, Qing Wang*, Peiliang Wu, Hao Yang, Yanchen Huo, Lei Yang, Li Wang, Yadan Luo, and Zheng Zhu

Abstract—World models predict environmental changes to help agents anticipate what may happen next and what different actions may lead to. Joint-Embedding Predictive Architectures (JEPAs) predict feature representations of observations without reconstructing complete images or videos, offering an approach to learning such predictive models. Initially developed for self-supervised representation learning, JEPA has gradually extended to tasks involving environmental change and action decisions. This survey reviews that development and organizes existing methods by their primary prediction tasks into five families: foundational or static, temporal, action-conditioned, multimodal, and structured or scene-centric JEPA. We further examine representative methods at four capability levels—representation, prediction, intervention, and control (R/P/I/C)—distinguishing the goals of learning useful features, predicting the future, capturing action effects, and supporting control. We also introduce commonly used datasets and evaluation metrics, compare reported performance, review major applications, and discuss current challenges and future research directions.

Index Terms—Joint-Embedding Predictive Architectures (JEPA), World Models, Predictive Representation Learning, Self-Supervised Learning.

1 Introduction

Before picking up an object, a robot needs to determine where it is and how it may move during the grasp. Before changing lanes, an autonomous vehicle needs to estimate the motion of nearby vehicles and the effects of its own action [1], [2]. Both tasks require an agent to predict environmental changes from current observations. World models address this problem by learning how environments evolve, allowing agents to compare possible outcomes before acting and revise their decisions as new observations arrive [3], [4], [5].

How to represent and predict the environment is a basic design question for world models. Some methods directly generate future images or videos, making their predictions easy to inspect [3], [6], [7]. However, a complete image contains colors, textures, and background details that may have little influence on the current decision. Other methods convert observations into compact feature representations and predict how these features change [8], [9]. These representations, often called latent states, aim to retain information needed for prediction and decision making while reducing the need to model every visual detail [4], [10], [11].

Joint-Embedding Predictive Architectures (JEPAs) build on predictive representation learning [12], [13] and learn by predicting features. For an image, a model uses visible regions to predict the features of hidden regions instead of recovering every pixel [1], [14]. The prediction task can be constructed from the image itself, without a manually assigned category label for every image. Such self-supervised tasks allow models to learn representations from relationships within the observations [15], [16], [17], [18], [19], [20].

The scope of JEPA research has since expanded. Moving from individual images to video introduces object motion and scene changes into the prediction task [21], [22]. Adding actions allows a model to estimate the state that may follow their execution. Other studies combine images, sound, and language, or represent scenes through objects and their relations. These methods share the idea of representation prediction, but differ in what they predict and which tasks they serve.

Understanding this development toward world models requires a clear account of what prediction means in each study. Predicting missing image features from their context primarily serves representation learning. Predicting future states from past observations requires learning temporal change. Comparing the consequences of different actions also requires connecting actions to state changes. Using these predictions to control a robot or vehicle further involves action selection and execution feedback. These goals are related, and each requires experimental evaluation suited to the capability being studied.

Existing surveys offer different entry points into this literature. Surveys of visual self-supervised learning describe how features can be learned from unlabeled data and how they perform on downstream tasks [23]. World-model surveys cover environmental prediction and model-based decision making [24], [25], while JEPA-focused reviews discuss learning approaches, architectures, and applications [26], [27]. This survey brings together JEPA research from these fields, tracing how its prediction tasks have changed and how these changes bring the methods closer to world modeling. Table 1 compares the scope of related surveys.

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Table 1. Comparison of related surveys by coverage.

Survey	Focus	JEPA method taxonomy	World-model capabilities	Action and control	Evaluation protocols	Quantitative results
Jing and Tian [23]	Visual SSL	✗	✗	✗	✓	✓
Khan *et al*. [28]	Vision Transformers	✗	✗	✗	◐	✓
Baltrušaitis *et al*. [29]	Multimodal learning	✗	✗	✗	◐	✗
Guan *et al*. [30]	Driving world models	✗	✓	✓	◐	✗
Ding *et al*. [25]	General world models	✗	✓	◐	◐	✗
Brotee *et al*. [26]	JEPA and world models	✓	✓	◐	◐	◐
Li *et al*. [31]	Embodied world models	✗	✓	✓	✓	✓
Wang *et al*. [32]	World-action models	✗	✓	✓	✓	✗
Lu *et al*. [33]	Robotic world-action models	✗	✓	✓	✓	✗
Jena *et al*. [34]	WM benchmarks	✗	◐	◐	✓	✗
This survey	JEPA-based world models	✓	✓	✓	✓	✓

✓ substantial coverage; ◐ related discussion; ✗ outside the principal scope. Quantitative results refer to reported model performance, not publication or benchmark counts. Symbols indicate coverage, not survey quality.

We organize existing methods by their primary prediction tasks into five families: foundational or static JEPA (JEPA-F), temporal JEPA (JEPA-T), action-conditioned JEPA (JEPA-A), multimodal JEPA (JEPA-M), and structured or scene-centric JEPA (JEPA-SC). This taxonomy organizes the different technical approaches and guides our review of their inputs, prediction targets, and main designs. Figure 1 shows representative developments.

Alongside this taxonomy, we use four capability levels—representation, prediction, intervention, and control (R/P/I/C)—to analyze experimental results. We ask four questions: Can the learned features support understanding of the environment? Can the model predict future states? Can it distinguish changes caused by different actions? Can its predictions help a system select actions and complete closed-loop tasks? These questions help identify the capabilities already demonstrated by each family and the gaps that remain relative to its intended applications.

The main contributions of this survey are as follows:

- We review JEPA research from representation learning toward world models, organize it into five method families, and describe their main ideas and representative studies.
- We analyze reported experiments at the R/P/I/C capability levels to assess which capabilities the reviewed methods possess.
- We summarize commonly used datasets, benchmark tasks, and evaluation metrics, and compare the reported performance of representative JEPA methods and related baselines.
- We review the main applications of JEPA and discuss current challenges and future research directions.

Section 2 introduces how JEPA learns and defines the R/P/I/C capability levels. Section 3 reviews the five method families and their technical development. Section 4 presents evaluation benchmarks, metrics, and performance results. Section 5 discusses major applications, and Section 6 examines current challenges and future directions. The concluding section summarizes the survey.

2 Foundations: JEPA Latent Prediction and World-Model Capabilities

JEPA uses available information to predict feature representations of target content without reconstructing the original observations. This section introduces its basic learning process and training objective, and explains its relation to world models through representation, prediction, intervention, and control. We then compare JEPA with related predictive and generative learning approaches.

2.1 JEPA Representations and Predictive Learning

Let $x_{\mathcal{C}}$ denote the available context and $x_{\mathcal{T}}$ a target selected by a sampling rule. Context and target may be spatial regions, temporally separated observations, or signals from different modalities. A common JEPA formulation encodes them separately and predicts the target representation from the context:

$$z_{\mathcal{C}} = E_\theta(x_{\mathcal{C}}), \tag{1}$$

$$z^*_{\mathcal{T}} = E_{\bar{\theta}}(x_{\mathcal{T}}), \tag{2}$$

$$\hat{z}_{\mathcal{T}} = P_\phi(z_{\mathcal{C}}, q_{\mathcal{T}}), \tag{3}$$

where $q_{\mathcal{T}}$ supplies target information such as location, time, or other conditioning. These equations describe the context–target prediction relation, not a fixed three-branch architecture shared by all JEPA methods. The target sampling rule and conditioning determine which relation the predictor learns.

A common training form minimizes the discrepancy between predicted and target representations:

$$\mathcal{L}_{\text{JEPA}} = D(\hat{z}_{\mathcal{T}}, \text{sg}[z^*_{\mathcal{T}}]) + \lambda\mathcal{R}, \tag{4}$$

where D measures latent prediction error, sg denotes a stop-gradient operation in this form, and $\mathcal{R}$ constrains the representation against collapse or poorly conditioned geometry [1], [13], [14], [35]. Target-encoder updates differ across methods; some variants simplify the interface or replace predictor–target alignment with direct representation alignment and regularization. Eq. (4) summarizes a common objective rather than every implementation.

The loss acts in representation space, so reconstructing $x_{\mathcal{T}}$ is unnecessary. Whether the resulting features retain information useful for future decisions also depends on target selection and on what the target encoder preserves. A successful latent prediction loss alone therefore makes no claim about world-model capability.

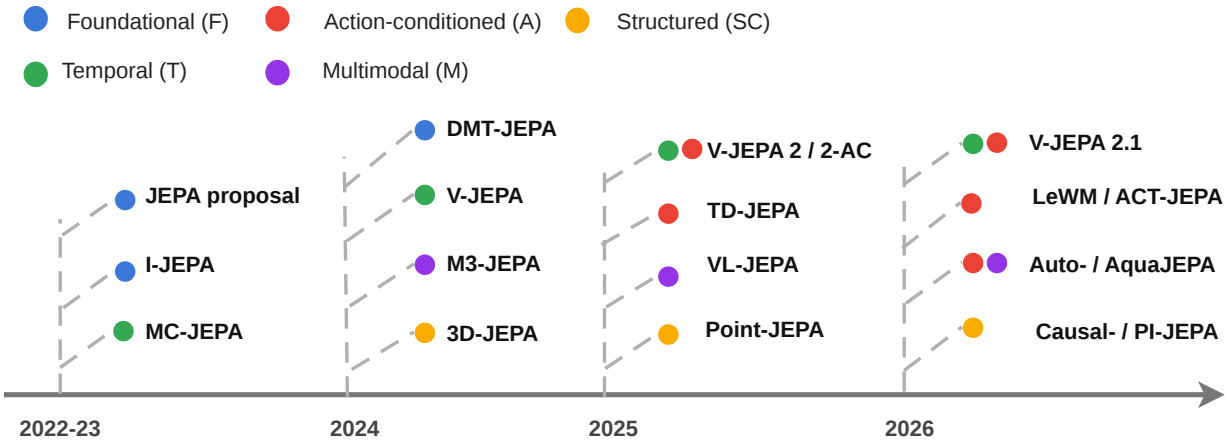


Fig. 1: Representative work in the five JEPA method families. The timeline shows representative developments, without indicating the number of papers in each family or ranking the families.

2.1.1 *Representation Stability and State Sufficiency*

A predictor can obtain a small loss if distinct inputs collapse to similar or low-rank embeddings. Stop-gradient, moving-average targets, and other mechanisms used in self-supervised learning address different routes to collapse [36], [37], [38], [39], [40], [41], [42], [43]. Statistical constraints offer a complementary approach: VICReg constrains variance and covariance, while SIGReg and related methods encourage well-conditioned representation geometry [13], [35], [44], [45], [46].

Preventing collapse and learning a sufficient state are different requirements. An embedding may separate objects well enough for recognition yet omit velocity or contact information needed to forecast their motion. Analyses of slow-feature bias, auxiliary targets, reconstruction, and representation geometry further show that a stable embedding need not preserve every variable relevant to future prediction [47], [48], [49], [50], [51], [52], [53], [54]. Stability concerns whether a representation remains informative; state sufficiency concerns whether it retains the information required by a specified prediction or decision task. The latter provides the link from JEPA representation learning to world modeling.

2.2 JEPA-Based World Modeling: Definition and Capabilities

In this survey, world modeling concerns using available observations to form an internal state and predict task-relevant environmental change. Prediction may occur in observation or latent space. Action sensitivity and closed-loop use require further evidence beyond the presence of a predictive training objective.

2.2.1 *A Latent State-Space View*

A controlled, partially observed environment can be written as

$$s_{t+1} \sim p(s_{t+1} \mid s_t, a_t), \tag{5}$$
$$o_t \sim p(o_t \mid s_t), \tag{6}$$

where s_t is environmental state, a_t an action, and o_t an observation. The first relation describes state evolution; the second describes how observations arise from state. The environmental state s_t is a conceptual variable and is not identified with a JEPA embedding.

A Markov state contains the historical information needed to predict the next transition:

$$p(s_{t+1} \mid s_{\le t}, a_{\le t}) = p(s_{t+1} \mid s_t, a_t). \tag{7}$$

Under partial observability, one observation may not reveal velocity, contact mode, or other variables needed for prediction. We denote the available observation–action history by

$$h_t = (o_{\le t}, a_{<t}). \tag{8}$$

A state-space reading of JEPA treats the encoder as estimating a useful latent state from available information and the predictor as modeling its evolution:

$$z_t = E_\theta(h_t), \tag{9}$$
$$\hat{z}_{t+h} = P_\phi(z_t, a_{t:t+h-1}). \tag{10}$$

This is an analytical correspondence, not a claim that every JEPA encodes full histories or accepts actions. Nor must z_t recover the complete physical state. It needs to retain information required for the future variables under study.

Let $Y_{t+1:t+H}$ denote task-relevant future variables over horizon H. An ideally predictively sufficient latent state satisfies

$$p(Y_{t+1:t+H} \mid h_t, a_{t:t+H-1}) = p(Y_{t+1:t+H} \mid z_t, a_{t:t+H-1}). \tag{11}$$

Given the considered actions, z_t then carries the information in the history needed to predict the declared future variables. This predictive-state criterion [55], [56] is idealized; minimizing a JEPA loss does not guarantee it. Sufficiency depends on both task and horizon: features adequate for recognition may omit pose or velocity needed for forecasting.

2.2.2 Representation, Prediction, Intervention, and Control

The state-space view leads to four levels for reading experimental evidence (Fig. 2). They describe the capabilities tested, not a sequence of modules that every JEPA must implement.

Representation (R). R asks whether the latent state retains information needed for the declared task. Downstream readouts can show whether particular variables are recoverable; the absence of collapse alone cannot. Predictive sufficiency in Eq. (11) is a stronger, task-dependent ideal.

Prediction (P). P asks whether the learned predictor forecasts an unobserved future beyond the available temporal context. Predicting a masked region or omitted content within an observed interval can support representation learning without testing future-state prediction. A generic autonomous transition is

$$\hat{z}_{t+h} = P_\phi(z_{\le t}), \qquad h > 0. \tag{12}$$

Evidence for P requires evaluation on the future relation. A one-step result does not establish direct multi-step prediction or stable recursive long-horizon rollout.

Intervention (I). I asks whether changing an action changes the predicted future appropriately. An action-conditioned interface is

$$\hat{z}_{t+h} = P_\phi(z_{\le t}, a_{t:t+h-1}), \tag{13}$$

but accepting $a_{t:t+h-1}$ is not evidence that the predictor has learned action effects. Evidence requires comparing supported alternative actions and checking whether the predicted differences correspond to their consequences.

Control (C). C asks whether learned representations and predictions improve decisions in a system that repeatedly receives environmental feedback. We distinguish two claims:

1) System-level control, C_{sys}, asks whether the complete system using the learned model achieves successful closed-loop behavior. It does not identify which component produces the gain.
2) Prediction-attributable control, C_{attr}, asks whether access to the learned predictive interface improves decisions. This requires a matched closed-loop comparison that holds the task, planner, observations, feedback schedule, and decision budget fixed while removing, bypassing, randomizing, or replacing the learned rollout.

Within a valid closed-loop setting, C_{attr} makes a stronger attribution claim than C_{sys}. The R/P/I/C levels describe progressively stronger evidence; they neither prescribe a fixed architecture nor follow directly from a method's taxonomy label.

In Table 2, D marks a direct test of the stated capability, Q marks related but incomplete evidence, and – means that the original study does not report a sufficient test. A – does not mean the model lacks that capability. For I, we look for a comparison of outcomes under different actions; for C_{attr}, we look for a matched closed-loop comparison with and without the predictive component.

2.3 Predictive and Generative Learning Paradigms

Predictive and generative learning paradigms differ in their targets, the spaces in which losses are computed, and the outputs their learned models expose. Reconstruction-based methods predict observation-level content, making output detail directly inspectable [19]. Contrastive and non-contrastive methods learn representations through relations among views and alignment objectives [18], [41], [42], [57]; their objectives need not provide a future-state predictor. CPC and data2vec move supervision into representation space by predicting future or teacher-generated features [12], [58], [59]. JEPA follows this latent-prediction direction, while the target construction and conditioning determine the relation learned by a particular method [1], [14].

Autoregressive models factor conditional distributions through sequential prediction [60], whereas diffusion models learn generative distributions through iterative denoising [61]. Both can provide detailed, inspectable outputs. JEPA primarily predicts in a learned latent space, whose utility depends on whether target representations retain variables needed by subsequent tasks. These mechanisms are not exclusive: latent prediction, sequential modeling, and generative decoding can serve different roles in one system. Output format alone does not establish a hierarchy of world-model capabilities.

We therefore compare paradigms by their targets and predictive interfaces, and assess capability claims using the corresponding R/P/I/C evidence. The taxonomy in Sec. 3 organizes JEPA methods by prediction task; the following performance section examines what their reported evaluations support.

3 JEPA Taxonomy and Technical Evolution

This section groups JEPA methods into five families according to their prediction tasks. JEPA-F studies representation prediction within static observations; JEPA-T addresses motion and temporal change in video; JEPA-A links latent prediction to action learning or action-conditioned state transitions; JEPA-M combines information such as images, sound, and language for prediction; and JEPA-SC focuses on objects, geometric structure, and their relations. Figure 3 illustrates the basic ideas of these five families.

We review representative work in each family, describing the inputs, prediction targets, model designs, and training approaches. Table 2 summarizes assessments of these methods in representation, prediction, intervention, and control.

Several distinctions in Table 2 are worth noting. Discrete-JEPA's P rating rests on a separate autoregressive world model trained on its discrete representations, rather than on the tokenizer alone [62]. ACT-JEPA reports policy success in simulated tasks, which supports C_{sys}, but its latent predictor does not take candidate actions as input [70]. No listed method receives D for C_{attr}: although several report closed-loop results, the reported comparisons do not fully isolate the gain from the learned predictor under otherwise matched conditions.

3.1 JEPA-F (Foundational Static Representation)

JEPA-F draws context and targets from a single observation, without requiring a model of how the environment changes over time. Its basic question is whether predicting a missing region's latent representation can produce transferable visual features [1], [14]. Methods in this family differ first in how they select and construct the target.

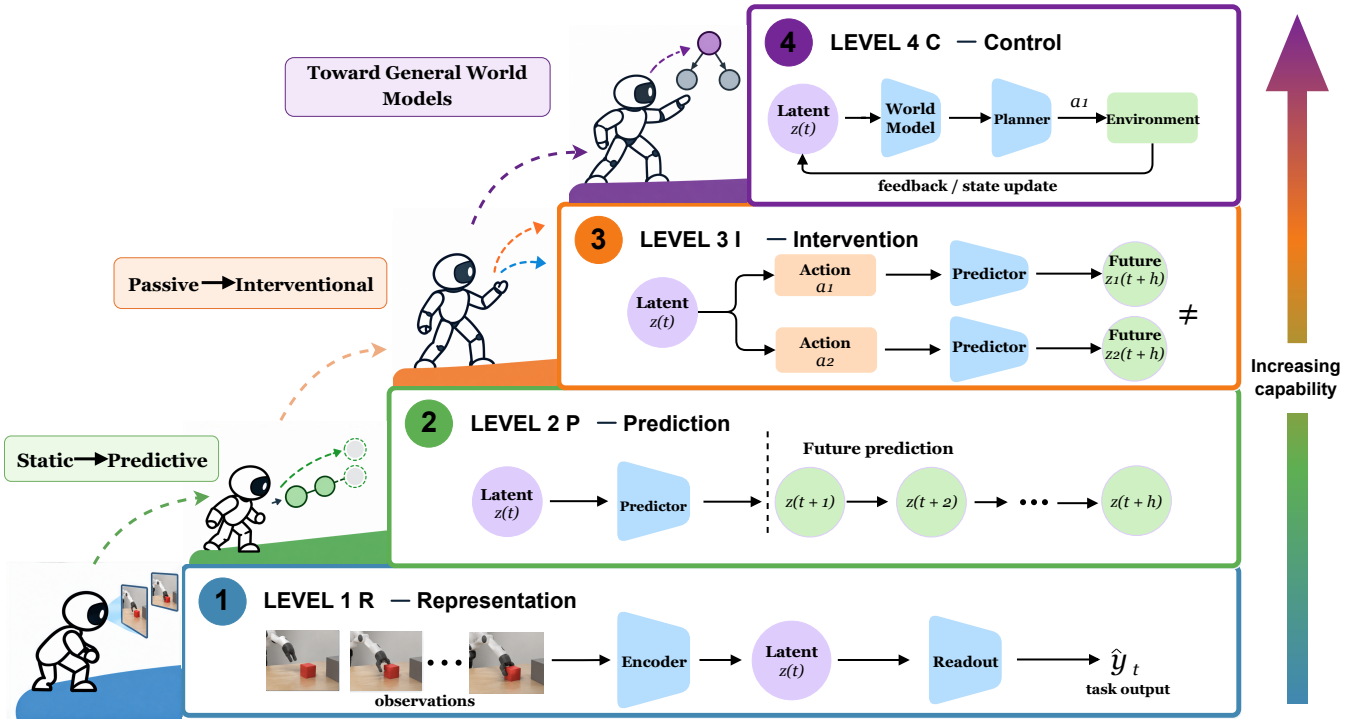


Fig. 2: A progressive capability hierarchy for JEPA-based world models. Representation, prediction, intervention, and control correspond to progressively stronger world-model capabilities.

3.1.1 *Masked-Region Latent Prediction*

I-JEPA selects a visible context block and several target blocks from one image [14]. A context encoder processes the visible region, a target encoder produces the training targets, and a predictor uses contextual and target-location information to estimate their representations. The model need not recover masked pixels. The scale of the target blocks and the information available in the context affect whether the task encourages semantic features. Figure 4 illustrates this formulation.

3.1.2 *Targets for Local Semantics*

Predicting masked block embeddings may still fail to retain the local information needed for dense tasks. DMT-JEPA constructs discriminative targets from semantically related neighboring patches [63]. Its distinction from I-JEPA lies chiefly in how the target is formed, rather than in whether prediction occurs in latent space. This difference helps explain their behavior on classification, segmentation, and detection tasks.

3.1.3 *Constraints on Representation Space*

Other work addresses collapsed or poorly conditioned embeddings. Sparse-JEPA introduces structured sparsity [64]. LeJEPA studies regularization of representation geometry, while VICReg provides a related self-supervised reference for preventing collapse [13], [35]. Such constraints shape the representation but do not, by themselves, establish how much information it retains for future prediction.

JEPA-F establishes the basic practice of predicting target representations from context and shows the importance of target design and representation constraints. Prediction within a single image mainly tests static features; environmental change requires temporal evaluation.

3.2 JEPA-T (Temporal JEPA)

JEPA-T places context and targets in video sequences. A predictor may then need motion and event-order information in addition to spatial cues. Video training alone, however, does not show that a model predicts an unobserved future: masked video content and future states define different prediction tasks.

3.2.1 *Masked Spatiotemporal Representation Prediction*

V-JEPA predicts masked spatiotemporal representations from visible video context [21]. Its targets come from the training video, but prediction across frames can yield features useful for motion understanding. MC-JEPA combines optical-flow estimation and semantic representation learning in a shared framework, allowing motion and visual content to inform one another [66]. These methods chiefly test whether predictive video learning produces useful temporal features.

3.2.2 *Future-Oriented Latent Prediction*

Another line moves the target toward future states. V-JEPA 2 connects video pretraining to action anticipation and tasks involving physical interaction [22]. CF-JEPA uses mask-free, multi-horizon forward prediction to model the evolution of latent states more explicitly [67]. Figure 5 illustrates the temporal context–target relation in V-JEPA 2 [22].

The two lines require different readings of their evaluations. Video recognition and action anticipation can show that

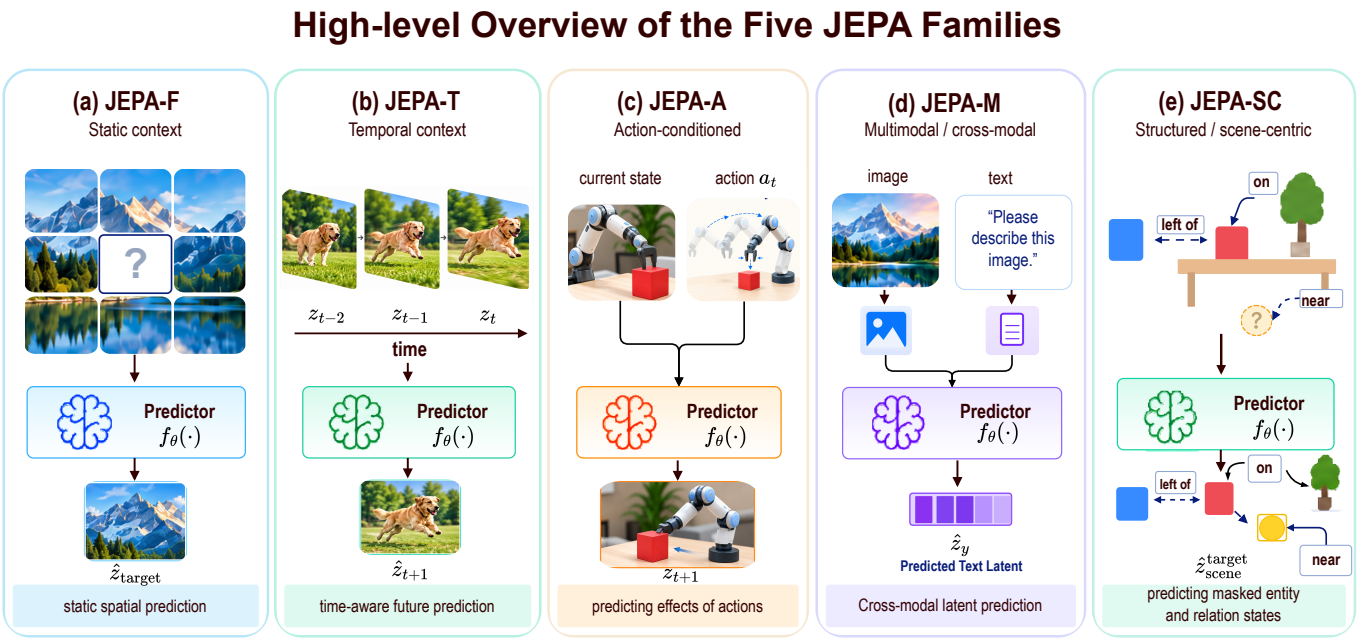


Fig. 3: Overview of the five JEPA families in this survey: (a) JEPA-F, static contextual latent prediction; (b) JEPA-T, temporal and future-oriented latent prediction; (c) JEPA-A, action-conditioned future-state prediction; (d) JEPA-M, cross-modal latent prediction; (e) JEPA-SC, structured scene-centric prediction of entities and relations. The schematics show family-level predictive patterns, not exact individual architectures.

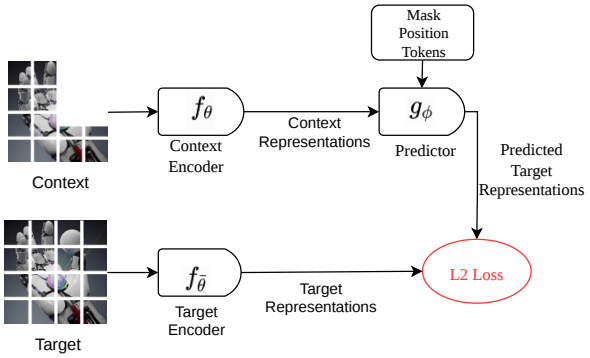


Fig. 4: Masked-current latent prediction in JEPA-F, illustrated with I-JEPA, where visible image regions are used to predict latent representations of masked target regions. Schematic redrawn and adapted from [14].

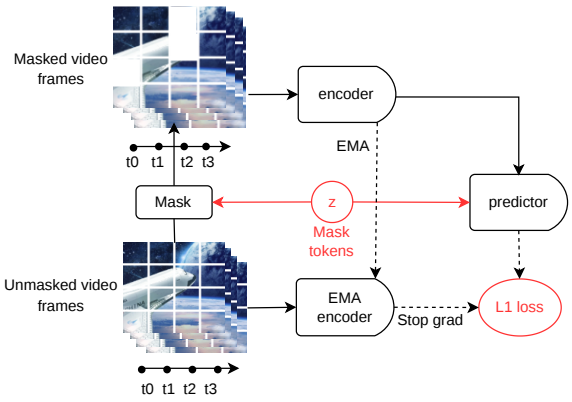


Fig. 5: Temporal latent prediction in JEPA-T, illustrated with V-JEPA 2, where temporal context is used to predict target representations along the temporal dimension. Schematic redrawn and adapted from [22].

features carry temporal information. Continuous prediction of future states requires direct tests of the predicted states and their stability as the horizon grows.

3.3 JEPA-A (Action-Conditioned JEPA)

Temporal prediction can model how an environment changes, but an agent also needs to estimate what its actions may cause. Most JEPA-A methods condition future latent-state prediction on candidate actions; others jointly learn latent prediction and an action policy. For action-conditioned predictors, the key question is whether different actions produce appropriately different forecasts.

3.3.1 Latent Prediction and Action Learning

V-JEPA 2-AC trains an action-conditioned predictor on robot trajectories while keeping a pretrained visual encoder fixed [22]. Visual representations and actions jointly condition

Table 2. JEPA families and their reported capability evidence. **D**: direct test; **Q**: related but incomplete test; **–**: no sufficient reported test.

Family / System	R	P	I	C_{sys}	C_{attr}
JEPA-F					
I-JEPA [14]	D	–	–	–	–
Discrete-JEPA [62]	D	D	–	–	–
DMT-JEPA [63]	D	–	–	–	–
SparseJEPA [64]	D	–	–	–	–
T-JEPA [65]	D	–	–	–	–
JEPA-T					
V-JEPA [21]	D	–	–	–	–
V-JEPA 2 [22]	D	Q	–	–	–
MC-JEPA [66]	D	–	–	–	–
CF-JEPA [67]	D	D	–	–	–
Branch-JEPA [68]	Q	D	–	–	–
UWM-JEPA [69]	D	D	D	–	–
JEPA-A					
V-JEPA 2-AC [22]	D	D	Q	D	Q
ACT-JEPA [70]	D	D	–	D	–
LeWM [71]	D	D	Q	D	Q
Fast-LeWM [72]	D	D	Q	D	Q
Sub-JEPA [73]	D	D	Q	D	Q
JEPA-M					
VL-JEPA [74]	D	–	–	–	–
M3-JEPA [75]	D	–	–	–	–
Le MuMo JEPA [76]	D	–	–	–	–
MJEPA [77]	D	–	–	–	–
AV-JEPA [78]	D	–	–	–	–
AquaJEPA [79]	D	D	D	D	Q
JEPA-SC					
Point-JEPA [80]	D	–	–	–	–
Causal-JEPA [81]	Q	D	Q	D	Q
3D-JEPA [82]	D	–	–	–	–
Cross4D-JEPA [83]	D	–	–	–	–

future latent-state estimates, providing an interface for planning. ACT-JEPA jointly learns latent future-state prediction and action sequences through imitation learning; its latent predictor does not use candidate actions as input [70]. It therefore connects prediction to action learning without directly testing the action-conditioned transitions studied by V-JEPA 2-AC. Figure 6 depicts V-JEPA 2-AC.

3.3.2 *Latent Rollouts for Planning*

LeWM jointly learns visual representations and latent dynamics from pixels, then uses predicted latent trajectories to select actions [71]. Fast-LeWM reduces the computational cost of sequential prediction for planning [72]. Sub-JEPA studies alternative latent learning strategies for more stable predictive models [73]. These methods place predicted trajectories in the decision loop, extending evaluation beyond features or isolated predictions.

Action inputs and task success need separate interpretation. A model may exploit correlations between actions and observations without correctly predicting the consequences of a changed action. Closed-loop success also depends on the planner and feedback. Testing action modeling therefore requires comparing predicted futures for alternative actions; attributing control gains to prediction requires a matched planning comparison.

3.4 JEPA-M (Multimodal JEPA)

An environment may be observed through images, audio, language, and other sensors. JEPA-M studies how latent prediction can use these heterogeneous sources. Current

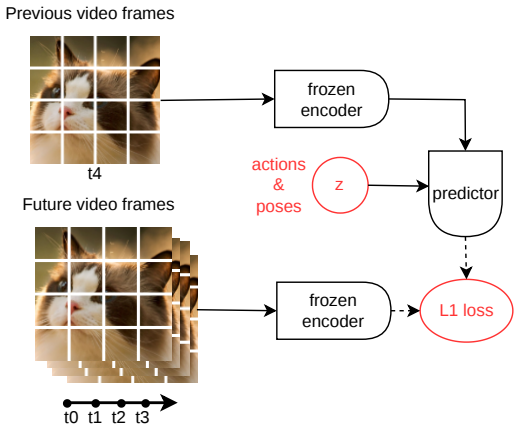


Fig. 6: Action-conditioned latent prediction in JEPA-A, illustrated with V-JEPA 2-AC, where visual representations and action-related inputs are used to predict future latent states. Schematic redrawn and adapted from [22].

methods mainly predict a target in one modality from another, or fuse information from several modalities before prediction.

3.4.1 *Cross-Modal Target Prediction*

VL-JEPA predicts continuous text-side embeddings from visual context, rather than using autoregressive token generation as its primary training objective [74]. It tests whether visual information can correspond to language semantics in representation space. Figure 7 shows this relation. Retrieval and classification can evaluate cross-modal correspondence, but do not themselves require prediction of future environmental change.

3.4.2 *Multimodal Fusion*

Fusion methods combine information from multiple modalities in a representation or prediction process. M3-JEPA uses a gated mixture-of-experts predictor to adjust how modalities are combined in latent space [75]. Other audio-visual and multi-sensor JEPA variants examine whether different input streams provide complementary cues [77], [78]. Their goal extends beyond a one-way mapping between two representations to using several inputs for prediction.

Multimodal world modeling calls for a further test of that complementarity. When an input is missing, sensory streams are asynchronous, or the target lies in the future, does the added modality still improve prediction? Static retrieval results cannot answer this question alone.

3.5 JEPA-SC (Structured / Scene JEPA)

A single global embedding can support recognition while omitting object positions, geometry, or relations between entities. JEPA-SC instead organizes targets around points, parts, objects, or trajectories. Existing work can be understood through two lines: geometric structure and entity relations.

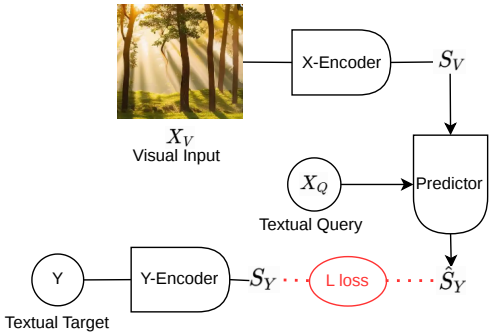


Fig. 7: Cross-modal target prediction in JEPA-M, illustrated with VL-JEPA, where visual context is used to predict a language-side latent target. Schematic redrawn and adapted from [74].

3.5.1 Point Clouds and Spatiotemporal Geometry

Point-JEPA and 3D-JEPA apply latent prediction to point clouds and three-dimensional representations, seeking to retain geometry without fully reconstructing observations. Cross4D-JEPA uses cross-modal correspondences to study spatiotemporal geometric change in dynamic scenes [80], [82], [83]. Their targets contain spatial structure useful for object or part understanding, rather than only global scene semantics.

3.5.2 Objects and Relations

Entity-centered methods separate a scene into objects and their relations. Causal-JEPA masks object tracks and predicts their latent states from the remaining entities and optional actions [81]. Figure 8 shows this structured prediction. It makes interacting entities explicit in the prediction task, but object-level prediction and relational modeling alone do not determine the direction of causal influence.

JEPA-SC shows how latent prediction can retain more explicit scene structure. Whether that structure also supports state transitions, action effects, or control must be judged from temporal and interactive experiments.

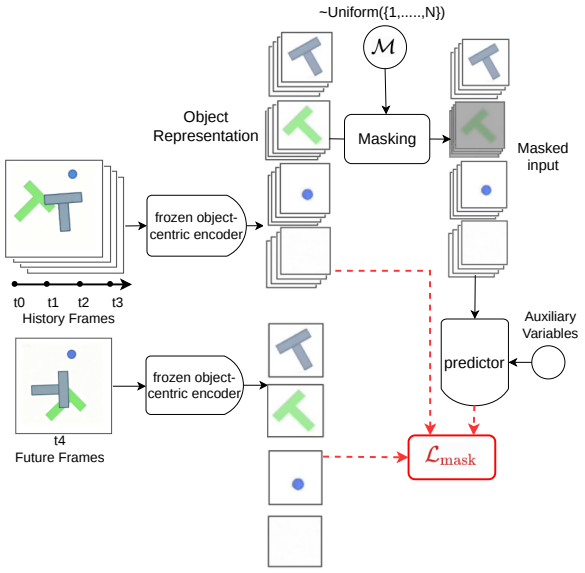


Fig. 8: Entity-structured target prediction in JEPA-SC, illustrated with Causal-JEPA, where structured relational context is used to predict masked object-level latent states. Schematic redrawn and adapted from [81].

Table 3. Benchmark tasks and metrics in the five performance tables.

Benchmark / task	Metric
JEPA-F	
IN-1K [93]	Lin. acc.; FT acc.; 1% acc.
ADE20K [94]	mIoU
COCO [95]	AP_{box}; AP_{mask}
DAVIS-17 [96]	J&F; $\mathcal{J}$; $\mathcal{F}$
JEPA-T	
SSv2 [97]; K400 [98]	Top-1 acc.
EK100 [99]	Act. mcR@5
JEPA-A	
2-Room; Reacher; PushT; Cube [72]	Success (%); Avg. (%)
JEPA-M	
COCO [95]	I→T R@1; T→I R@1
Video–language [74]	Avg. R@1
Video–language [74]	Avg. Cls. Acc.
AS-20K [100]	A+V mAP
JEPA-SC	
ScanObjectNN PB-T50-RS [101]	OA
ShapeNetPart [102]	$mIoU_I$

4 Performance Evaluation

This section introduces the datasets and evaluation metrics commonly used for the five JEPA families and summarizes the main performance patterns in Tables 4–8.

4.1 Datasets and Evaluation Metrics

Table 3 summarizes the tasks and metrics used in the following performance tables. Results are taken from the original studies. Scores are directly comparable only when the task, data split, and evaluation protocol match; otherwise, they show each method's performance under its reported setting. Differences in backbones [84], [85], [86], [87], [88] and pretraining [89], [90], [91], [92] also affect the comparison.

4.1.1 Datasets and Tasks

Tables 4–8 cover image understanding, video understanding, action-conditioned control, multimodal learning, and 3D perception. The datasets and tasks are described below.

(1) Image classification and dense prediction

ImageNet-1K (IN-1K) contains 1,000 object categories and is commonly used for image classification. In addition to evaluation with all training labels, limited-label settings test how well pretrained models perform when annotations are scarce [93].

ADE20K provides pixel-level semantic annotations for indoor and outdoor scenes, with 150 categories in the standard scene-parsing setting. Models assign a category to each pixel, separating objects and background regions within an image [94].

COCO provides object categories, bounding boxes, and instance masks. Object detection requires recognizing and locating objects, while instance segmentation also requires predicting the pixels belonging to each object, placing greater demands on spatial detail [95].

DAVIS-2017 (DAVIS-17) provides videos with frame-wise object masks. The mask-propagation task in Table 4 uses available annotations to locate the same object in later frames, testing feature correspondence under motion, deformation, and occlusion [96].

(2) Video recognition and action anticipation

These benchmarks are commonly used to evaluate video representations [103], [104], [105], [106], [107].

Something-Something V2 (SSv2) focuses on everyday human–object interactions. Actions may involve similar objects but differ in motion direction or operation order, requiring models to use changes across successive frames [97].

Kinetics-400 (K400) contains 400 human action classes covering sports, daily activities, and human–object interactions. It covers a broader range of actions than SSv2, with recognition drawing on motion, object appearance, and scene information [98].

EPIC-KITCHENS-100 (EK100) records kitchen activities from an egocentric viewpoint and annotates actions with verbs, nouns, and their combinations. Table 5 uses action anticipation, requiring models to predict an upcoming action from the video observed so far [99].

(3) Action-conditioned planning and control

Task performance is evaluated through the resulting planner or policy [108], [109], [110], [111], [112].

Two-Room requires an agent to reach a goal through a passage connecting two rooms. The model predicts the effects of movement actions to help a planner select a feasible path.

Reacher requires an arm's end effector to reach a target position, testing predictions of subsequent motion from the current state and an action.

PushT requires pushing a T-shaped object to a target position and orientation. The outcome depends on the contact location and pushing direction, requiring predictions of object motion following contact.

Cube requires a sequence of manipulations to bring a block to a target state, testing whether predictions of manipulation outcomes can support subsequent action selection [72].

(4) Multimodal retrieval and classification

COCO image–text retrieval uses images and their associated descriptions. Image-to-text retrieval finds a description for an image, while text-to-image retrieval finds an image for a description, testing correspondence between visual content and language [75], [95].

Video–language benchmarks include video retrieval and classification. Retrieval finds videos from textual descriptions, while classification matches videos to category names or descriptions. Table 7 reports average results over multiple datasets [74].

The AS-20K setting of AudioSet is used for audio–visual event classification, with each clip potentially carrying multiple event labels. Models combine sound with visible objects and activities to identify the events present [100].

(5) 3D object recognition and part segmentation

ScanObjectNN contains real scanned object point clouds with background clutter, noise, and incomplete observations. Table 8 uses the PB-T50-RS setting to evaluate object classification under background and geometric perturbations [101].

ShapeNetPart provides point-wise part annotations, such as the back, seat, and legs of a chair. Models distinguish parts within an object, placing greater emphasis on local shape and part boundaries than whole-object classification [102].

4.1.2 *Evaluation Metrics and Settings*

Top-1 accuracy measures the proportion of samples whose highest-ranked predicted class matches the ground truth and is used for image classification and video action recognition. **Overall accuracy (OA)** in 3D object classification likewise measures the proportion of correctly classified test samples.

Intersection-over-union (IoU) measures the overlap between a predicted region and its annotation:

$$\mathrm{IoU} = \frac{|P \cap G|}{|P \cup G|},$$

where P and G denote the predicted and annotated regions, respectively. Greater agreement gives a higher IoU.

ADE20K uses **mean IoU (mIoU)**, averaged over semantic classes. ShapeNetPart uses **instance-averaged mIoU (mIoU$_I$)**: part IoUs are first averaged within each object, and the resulting scores are then averaged over object instances [102].

Average precision (AP) summarizes precision and recall as predictions are ranked by confidence. COCO averages AP over IoU thresholds from 0.50 to 0.95 in steps of 0.05, evaluating both object detection and localization quality. $\mathrm{AP}_{\mathrm{box}}$ uses bounding boxes, while $\mathrm{AP}_{\mathrm{mask}}$ uses pixel masks, for object detection and instance segmentation, respectively [95].

DAVIS-2017 region similarity $\mathcal{J}$ uses mask IoU to measure region overlap. **Boundary accuracy** $\mathcal{F}$ is an F-score computed from boundary-matching precision and recall, measuring contour quality. **J&F** is the average of the two [96].

EK100 uses **action mean-class top-five recall (mcR@5)**. Recall is computed for each action class as the proportion of samples whose true action appears among the top five predictions, then averaged over classes. Equal class weighting prevents frequent actions from dominating the overall score [99].

R@1 measures the proportion of queries for which a correct match ranks first. Image-to-text and text-to-image results are reported separately. In Table 7, Avg. R@1 averages scores over video retrieval datasets, while Avg. Cls. Acc. averages accuracy over video classification datasets.

Mean average precision (mAP) computes AP for each event class from confidence-ranked predictions, then averages over classes. It supports classification of clips containing multiple events. A+V in the table denotes the use of both audio and video inputs.

Success rate is the proportion of test episodes in which the task goal is reached. The **four-task average (Avg.)** in Table 6 is the arithmetic mean of success rates on Two-Room, Reacher, PushT, and Cube. Multi-seed experiments also report the mean and variation across runs.

Linear probing (Lin.) freezes the pretrained encoder and trains only a linear classifier. **Fine-tuning (FT)** updates some or all model parameters using downstream data. The

1% label setting uses only a small proportion of training annotations.

4.2 Performance Comparison

Tables 4–8 summarize representative results for the five JEPA families. Table 4 covers image classification, semantic segmentation, object detection, and video object segmentation. DMT-JEPA improves on I-JEPA across several tasks, while the rankings of different methods change between linear probing and fine-tuning, indicating that linear separability and downstream adaptability do not always favor the same representation. Table 5 reports video action recognition and action anticipation. V-JEPA 2.1-G outperforms InternVideo2 on SSv2 but is slightly lower on K400, reflecting different strengths in fine-grained interaction understanding and broader action recognition. Scaling V-JEPA 2.1 improves both recognition and anticipation performance [63], [113].

Table 6 compares success rates on navigation, reaching, object pushing, and cube manipulation. Fast-LeWM mainly improves Two-Room and Cube, while adding self-consistency further improves Reacher, PushT, and Cube, showing that the same model change can affect control tasks differently [72]. Table 7 summarizes image–text, video–language, and audio–visual tasks, where VL-JEPA BASE achieves the best results among the listed methods for zero-shot video–language retrieval and classification [74]. Table 8 reports 3D object classification and part segmentation. Longer 3D-JEPA pretraining improves both tasks, with a larger gain in object classification [82]. The tables cover five different task areas. The effects of model size and training time are best judged from configurations evaluated under comparable conditions within each task.

5 Application Domains

JEPA methods have been used in image and video understanding, 3D perception, robotics, autonomous driving, multimodal learning, and scientific data analysis. Their predictors play different roles across these settings: some applications use only the learned representations, others require estimates of future states, and robotics or driving systems may also use predictions to select actions. Table 9 lists representative methods and the capability requirements typical of each domain. Here R, P, I, and C denote representation, prediction, intervention, and control. These are requirements of the application, not capabilities established by every listed method.

5.1 Image and Spatial Understanding

Image classification, semantic segmentation, and object detection all require transferable visual features, but they depend on different information. Classification relies heavily on global semantics, whereas segmentation and detection also require object locations, boundaries, and local structure. I-JEPA learns visual features by predicting representations of masked image regions without reconstructing pixels [14]. DMT-JEPA and spatially conditioned variants further examine how local information can be retained in the representation [63], [144].

Such models can provide pretrained visual backbones for downstream tasks, particularly when unlabeled images are abundant and dense annotations are costly [145]. In recognition and segmentation, however, downstream inference usually uses the encoder output without running the predictor. These results therefore primarily test the utility of the representation; performance on static images alone does not establish that the model has learned environmental dynamics.

5.2 Video Understanding and Anticipation

Action recognition and event understanding require a model to use changes across video frames. V-JEPA, MC-JEPA, V-JEPA 2, and SALT use latent prediction during video pretraining to learn representations sensitive to motion and events without reconstructing every frame [21], [22], [66], [139]. In action recognition, prediction may remain a training mechanism, while the downstream model classifies actions already present in the observed clip from the learned video features.

Anticipating future events poses a different problem. Predicting a masked segment within an observed clip is not equivalent to predicting a state beyond the observation window. Results from V-JEPA 2 indicate that predictive video pretraining can capture information relevant to future events and physical interactions [22], [146]. Prediction over longer horizons must also account for accumulating errors and multiple plausible futures. Video applications should therefore report both the value of the representation for understanding tasks and whether the prediction target lies in the unobserved future.

5.3 3D and Structured Perception

Point-cloud recognition and scene understanding require representations that retain shape, position, and local geometry. Point-JEPA and 3D-JEPA apply joint-embedding prediction to point-cloud features; LiDAR-oriented work addresses sparse range observations; and Cross4D-JEPA extends prediction to cross-modal correspondences in dynamic 4D geometry [80], [82], [83], [147]. These methods seek to use unlabeled spatial data when annotations are limited or observations are incomplete.

Other studies shift the prediction target from a scene-level representation to entities and their relations. Causal-JEPA explores object-level latent-state prediction, while Social-JEPA models geometric structure among interacting entities [81], [148]. Such representations can identify which object or relation has changed, provided that entity separation and spatial correspondence remain stable. Point-cloud classification and segmentation test geometric features; assessing structured prediction also requires changes in viewpoint, occlusion, object composition, and interaction patterns.

5.4 Embodied Robotics

Robot applications connect visual representations to actions. In manipulation, a model may need to estimate how an object will change after a grasp or push; in navigation, it must use scene information to choose actions and update them as new observations arrive. V-JEPA 2-AC and LeWM

Table 4. JEPA-F representation performance across representative benchmarks.

(a) Image classification

Method	Paradigm	Backbone	PT Ep.	IN-1K Lin. ↑	IN-1K FT ↑	IN-1K 1% ↑
DINO [57], [63]	Self-distill.	ViT-B/16	400	78.2	82.8	–
MoCo v3 [63], [114]	Contrastive	ViT-B/16	300	76.5	–	–
BEiT [63], [115]	Masked image	ViT-B/16	800	56.7	83.4	–
iBOT [63], [116]	Self-distill.	ViT-B/16	400	79.5	84.0	–
MAE [19], [63]	Masked AE	ViT-B/16	1600	68.0	83.6	–
CAE [117]	Masked AE	ViT-B/16	1600	70.4	–	–
I-JEPA [14], [63]	JEPA	ViT-B/16	600	72.9	83.5	–
DMT-JEPA [63]	JEPA	ViT-B/16	600	73.8	84.6	–
VISReg [45]	JEPA	ViT-B/16	400	75.7	83.0	–
data2vec [59]	Predictive SSL	ViT-L/16	1600	77.3	–	73.3
MAE [19], [63]	Masked AE	ViT-L/16	1600	76.0	85.9	67.1
CAE [117]	Masked AE	ViT-L/16	1600	78.1	–	–
iBOT [116]	Self-distill.	ViT-L/16	250	81.0	–	–
I-JEPA [14], [63]	JEPA	ViT-L/16	600	77.5	–	69.4
DMT-JEPA [63]	JEPA	ViT-L/16	600	78.2	86.6	–
LeJEPA [35]	JEPA	ViT-L/14	100	75.6	–	–
MAE [19]	Masked AE	ViT-H/14	1600	77.2	–	71.5
MAE [19]	Masked AE	ViT-H/14 (448px)	1600	–	87.8	–
I-JEPA [14]	JEPA	ViT-H/14	300	79.3	–	73.3
I-JEPA [14]	JEPA	ViT-H/16 (448px)	300	81.1	87.1	77.3
DMT-JEPA [63]	JEPA	ViT-H/14	300	80.6	–	–

(b) Object detection and semantic/instance segmentation

Method	Paradigm	Backbone	PT Ep.	ADE20K mIoU ↑	COCO AP_{box} ↑	COCO AP_{mask} ↑
Supervised [63]	Supervised	ViT-B/16	–	47.4	47.9	42.9
DINO [57], [63]	Self-distill.	ViT-B/16	400	46.8	50.1	43.4
MoCo v3 [63], [114]	Contrastive	ViT-B/16	300	47.3	47.9	42.7
BEiT [63], [115]	Masked image	ViT-B/16	800	47.1	49.8	44.4
iBOT [63], [116]	Self-distill.	ViT-B/16	400	50.0	51.2	44.2
MAE [19], [63]	Masked AE	ViT-B/16	1600	48.1	50.3	44.9
I-JEPA [14], [63]	JEPA	ViT-B/16	600	47.6	49.9	44.5
DMT-JEPA [63]	JEPA	ViT-B/16	600	49.0	50.9	45.6
VISReg [45]	JEPA	ViT-B/16	400	30.2	–	–

(c) Video object segmentation

Method	Paradigm	Backbone	PT Ep.	DAVIS-17 J&F ↑	DAVIS-17 $\mathcal{J}$ ↑	DAVIS-17 $\mathcal{F}$ ↑
DINO [57], [63]	Self-distill.	ViT-B/16	400	62.3	60.7	63.9
MAE [19], [63]	Masked AE	ViT-B/16	1600	51.0	49.4	52.6
I-JEPA [14], [63]	JEPA	ViT-B/16	600	56.2	56.1	56.3
DMT-JEPA [63]	JEPA	ViT-B/16	600	57.7	56.7	58.7
MAE [19], [63]	Masked AE	ViT-L/16	1600	53.4	52.5	54.3
I-JEPA [14], [63]	JEPA	ViT-L/16	600	56.6	56.3	56.9
DMT-JEPA [63]	JEPA	ViT-L/16	600	58.3	57.3	59.2

investigate action-conditioned latent-state prediction, while TD-JEPA and Temporal-Distance JEPA study representations for decision making and planning [22], [71], [127], [149]. Related JEPA work also addresses cross-embodiment imitation, underwater robotics, and aerial sim-to-real transfer [79], [150], [151].

A latent predictor can let a planner compare the expected outcomes of candidate actions without generating a complete image for each one. Manipulation also depends on contact, slip, actuator delay, and partial observability, which may not be captured by representations that retain only object semantics. Non-JEPA visual representations, manipulation policies, and large demonstration datasets provide background and comparison points for this literature [152], [153], [154], [155], [156], [157], [158].

The experimental setting matters here: a simulated task, a real-robot trial, and a representation probe on robot data provide different evidence. Closed-loop task success reflects the combined perception, prediction, planning, and feedback system. To isolate the contribution of latent prediction, studies must compare systems with the predictor removed or replaced and examine whether the robot can replan after execution deviates from its prediction.

5.5 Autonomous Driving

Autonomous driving requires road-scene perception, forecasts of other road users, and timely decisions. Existing JEPA studies address different parts of this problem. V2X-JEPA uses information from multiple agents to learn features for cooperative perception, while LiDAR- and 4D-oriented methods provide spatial features of road scenes. Drive-JEPA uses video representations for trajectory learning, Auto-JEPA relates the predictive space to driving intent and trajectory variables, and HanoiWorld explores a latent JEPA-based world model for vehicle control [83], [140], [141], [147], [159], [160].

These tasks provide different levels of evidence. Cooperative-perception results concern the representation of road users; trajectory forecasting tests estimates of future motion; and control experiments test whether a model

Table 5. JEPA-T temporal representation and future prediction performance.

(a) Action recognition					
Method	**Paradigm / Setting**	**Backbone**	**Params.**	**SSv2 Top-1 ↑**	**K400 Top-1 ↑**
DINOv2 [113], [118]	Image SSL	ViT-g/14	1.1B	50.7	83.6
PE-Core G [113], [119]	V-L	ViT-G	1.9B	55.4	88.5
SigLIP2 [113], [120]	V-L	ViT-g	1.2B	49.9	87.3
InternVideo2$_{s2}$ [113], [121]	Video-L	ViT / video encoder	1.0B	69.7	89.4
V-JEPA [21], [113]	JEPA	ViT-H/16	0.6B	74.3	84.5
V-JEPA 2-g/384 [22], [113]	JEPA	ViT-g	1.0B	77.3	87.3
V-JEPA 2.1-g [113]	JEPA	ViT-g	1.0B	76.9	87.0
V-JEPA 2.1-G [113]	JEPA	ViT-G	2.0B	77.7	87.7
MJEPA [77]	JEPA / Audio-video	ViT-g	1.0B	73.9	85.0

(b) Action anticipation				
Method	**Paradigm / Setting**	**Backbone**	**Params.**	**EK100 Act. mcR@5 ↑**
V-JEPA 2-g/384 [22], [113]	JEPA	ViT-g	1.0B	39.7
V-JEPA 2.1-g [113]	JEPA	ViT-g	1.0B	38.4
V-JEPA 2.1-G [113]	JEPA	ViT-G	2.0B	40.8
InAViT [113], [122]	Supervised	Video Transformer	0.16B	25.8
Video-LLaMA [113], [123]	Video LLM	Video encoder + LLM	7B	26.0
PlausiVL [113], [124]	Video-L	Video-language encoder	8B	27.6

Table 6. JEPA-A planning and policy success under source-specific evaluation protocols.

(a) Four-task success and reported average							
Method	**Paradigm**	**Model / Backbone**	**2-Room Success (%) ↑**	**Reacher Success (%) ↑**	**PushT Success (%) ↑**	**Cube Success (%) ↑**	**Four-task Avg. Success (%) ↑**
PLDM [72], [125]	Latent WM	Latent dynamics model	97	78	78	65	79.5
DINO-WM [72], [126]	Feature WM	DINO visual encoder	100	79	74	86	84.8
LeWM [71]	JEPA-WM	Latent WM encoder	87	86	96	74	85.8
Fast-LeWM [72]	JEPA-WM	LeWM-based	98	88	96	80	90.5
Fast-LeWM+SC [72]	JEPA-WM	LeWM + SC	98	90	98	82	92.0

(b) Four-task multi-seed evaluation							
Method	**Paradigm**	**Model / Backbone**	**Seeds**	**2-Room Success (%) ↑**	**Reacher Success (%) ↑**	**PushT Success (%) ↑**	**Cube Success (%) ↑**
LeWM rerun [71], [73]	JEPA-WM	Latent WM encoder	6	84.33±4.23	82.67±4.42	84.67±6.53	67.33±5.01
Sub-JEPA [73]	JEPA-WM	JEPA latent predictor	6	95.00±2.76	84.00±4.00	89.00±5.33	76.33±5.99
LeWM [71], [127]	JEPA-WM	Latent WM encoder	10	97.4±1.3	96.0±1.9	83.6±3.2	68.0±2.8
RC-aux [127]	Aux. baseline	Auxiliary representation	10	98.6±1.0	96.8±1.4	81.4±1.9	81.6±2.8
TD-JEPA [127]	JEPA-WM	Temporal-distance predictor	10	100.0±0.0	97.0±2.4	86.0±4.2	82.2±2.9

(c) PushT-only evaluation			
Method	**Paradigm**	**Model / Backbone**	**PushT Success (%) ↑**
Causal-JEPA [81]	JEPA-WM / Object-centric	Object-centric predictor	88.67
ACT-JEPA [70]	JEPA / Imitation learning	JEPA encoder + action decoder	41.00

can support action under feedback. Driving also involves occlusion, road topology, multi-agent interaction, and rare but consequential events. A model may represent current traffic accurately yet miss an unusual maneuver by another vehicle, or fail to predict how a change in the ego vehicle's path would affect the scene. Evaluation for driving decisions should therefore specify action conditioning, prediction horizon, and closed-loop protocol rather than treating perception scores as evidence of prediction or control.

5.6 Vision-Language and Multimodal Agents

Multimodal JEPAs examine how different information sources contribute to a latent representation. VL-JEPA predicts language-side representations from visual context, and M3-JEPA explores multimodal latent alignment. Le MuMo JEPA, MJEPA, and AV-JEPA study different forms of multimodal fusion or audio-visual prediction [74], [75], [76], [77], [78]. VLA-JEPA and JEPA-VLA further connect visual predictive representations to vision-language-action systems [161], [162].

The applications differ in what they ask of the model. Cross-modal alignment tests whether images, sounds, and text have useful semantic correspondences. Multimodal state recognition tests whether an additional input resolves information missing from one modality. Interactive tasks require the model to predict state changes under instructions or actions. Retrieval or alignment results support the first of these claims, but do not establish the others. To assess predictive value, experiments should test missing or asynchronous modalities and changes in instructions, then determine whether the extra information improves subsequent actions.

5.7 Scientific and Sequential Domains

JEPA objectives have also been applied to brain activity, clinical records, medical images, and physical time series.

Table 7. JEPA-M cross-modal representation and transfer performance.

(a) Image–text retrieval

Method	Paradigm	Backbone / Params.	Setting	COCO I→T R@1 ↑	COCO T→I R@1 ↑
ALIGN [75], [128]	Dual encoder	EffNet-L2 + BERT-L	Fine-tuned	77.0	59.9
ALBEF [75], [129]	Align-and-fuse	ViT-B/16 + BERT-base	Fine-tuned	77.6	60.7
BLIP [75], [130]	V-L pretraining	ViT-L/16	Fine-tuned	82.4	65.1
BEiT-3 [75], [131]	Multimodal encoder	1.9B	Fine-tuned	84.8	67.2
BLIP-2-L [75], [132]	Frozen VLM	ViT-L/14 / 474M	Fine-tuned	83.5	66.3
BLIP-2-g [75], [132]	Frozen VLM	ViT-g/14 / 1.2B	Fine-tuned	85.4	68.3
M3-JEPA [75]	JEPA	≈8.5B total / 140M trainable	Fine-tuned (3-layer LoRA)	87.7	89.7

(b) Video–language evaluation

Method	Paradigm	Backbone / Params.	Setting	Video–Language Benchmarks Avg. R@1 ↑	Video–Language Benchmarks Avg. Cls. Acc. ↑
CLIP [74], [133]	Contrastive VLM	ViT-L / 389M	Zero-shot	35.3	30.7
SigLIP2 [74], [120]	Contrastive VLM	ViT-L / 882M	Zero-shot	45.4	38.6
SigLIP2 [74], [120]	Contrastive VLM	ViT-g / 1.9B	Zero-shot	47.5	39.8
PE-Core [74], [119]	V-L encoder	ViT-L / 671M	Zero-shot	50.2	42.9
PE-Core [74], [119]	V-L encoder	ViT-G / 2.3B	Zero-shot	58.1	44.7
VL-JEPA$_{\mathrm{BASE}}$ [74]	JEPA	ViT-L / 1.6B	Zero-shot	63.7	52.5

(c) Audio–visual classification

Method	Paradigm	Backbone / Params.	Setting	AS-20K A+V mAP ↑
MJEPA [77]	JEPA	ViT-g / 1.0B	Frozen	45.4
AV-JEPA [78]	JEPA	ViT-B/16	Fine-tuned	29.6

Table 8. JEPA-SC structured and geometric representation performance. ScanObjectNN uses the PB-T50-RS variant.

Method	Paradigm / Structure	Backbone	PT Epochs	ScanObjectNN OA ↑	ShapeNetPart mIoU$_I$ ↑
Transformer	Supervised	Point Transformer	–	79.11	–
Point-BERT [134]	Masked point	Point Transformer	300	83.07	85.60
Point-MAE [135]	Masked AE	Point Transformer	300	85.18	86.10
Point-M2AE [136]	Hierarchical AE	Hierarchical point Transformer	300	86.43	86.51
Point2Vec [137]	3D SSL	Point Transformer	800	87.50	86.30
ACT [138]	Cross-modal SSL	Point Transformer	300	88.21	86.14
Point-JEPA [80]	JEPA / Point cloud	Point Transformer	500	86.6 ± 0.3	85.8 ± 0.1
3D-JEPA [82]	JEPA / Point cloud	Point Transformer	150	88.65	86.28
3D-JEPA [82]	JEPA / Point cloud	Point Transformer	300	89.52	86.41

Table 9. Representative JEPA application domains and capabilities.

Application	Major capacity	Representative systems
Image understanding	R	I-JEPA [14], DMT-JEPA [63]
Video understanding	R, P	V-JEPA 2 [22], SALT [139]
Structured perception	R, P	Point-JEPA [80], Cross4D-JEPA [83]
Embodied robotics	R, P, I, C	V-JEPA 2-AC [22], LeWM [71]
Autonomous driving	R, P, I, C	Auto-JEPA [140], Drive-JEPA [141]
Multimodal agents	R, P, I	VL-JEPA [74], M3-JEPA [75]
Scientific domains	R, P	Brain-JEPA [142], Clin-JEPA [143]

Brain-JEPA learns representations of brain activity, Clin-JEPA processes longitudinal health records, and related studies apply predictive objectives to medical imaging [142], [143], [163], [164], [165]. In physical systems, Phys-JEPA and PI-JEPA explore multivariate state prediction and latent surrogate models, building on work in physical simulation, operator learning, and data-driven forecasting [166], [167], [168], [169], [170], [171]. Related predictive objectives have been studied for polymers, genomics, proteins, and crystal materials [172], [173], [174], [175].

These domains require more than low average prediction error. Clinical applications must consider reliability in rare states, while physical models should also be checked against known constraints. A latent representation that omits task-relevant variables may perform well on classification yet remain unsuitable for explaining system changes. Predictive representation learning on scientific data shows the reach of JEPA objectives; whether a system is an interactive world model depends on evidence for state evolution, external interventions, and their consequences.

6 Challenges and Future Directions

JEPA has extended from image representation learning to temporal prediction and action-conditioned modeling. For latent prediction to support reliable world models, several related questions remain: whether the representation retains the necessary state information, whether predictions reflect the effects of actions, and whether the model remains useful for decisions over longer horizons.

6.1 Representation Stability and Adaptive Prediction Targets

JEPA training commonly uses stop-gradient operations and an exponential-moving-average target encoder, with variance–covariance regularization in some formulations, to avoid representational collapse [13], [14], [41], [42], [44]. Yet a representation can avoid collapse without being suitable for later prediction. Features may become concentrated along

a small number of directions as effective dimensionality declines; even recognizable semantic features may be poorly conditioned for state transport or decoding [35], [45], [46], [176]. Low prediction loss likewise does not guarantee that the latent state contains variables needed for decisions.

Prediction targets must reflect the task. Image recognition benefits from invariance to irrelevant appearance changes, dense perception needs locations and boundaries, and control requires sensitivity to small changes that affect the next state. Multimodal masked-learning studies have already varied visible inputs and prediction targets across tasks [177]. For JEPA, the joint effects of masking, target granularity, and prediction horizon deserve closer study: a coarse target may favor semantics while losing local state, whereas a fine target may emphasize irrelevant detail. Adaptation to new tasks or sensors should therefore test whether the existing representation geometry survives, rather than relying only on accuracy on the new task.

6.2 Invariance, Action Grounding, and Counterfactual Sensitivity

Representation learning often suppresses changes in color or imaging conditions, but contact, pose, object identity, and action-dependent configuration can determine how an environment evolves [43], [47], [63], [178]. Suppressing these differences may leave recognition performance intact while preventing a model from distinguishing the consequences of alternative actions. Augmentations useful for natural images may also be unsuitable for physical systems when a seemingly small change carries information needed for prediction [179].

Passive video and trajectories collected by an existing policy make this distinction harder. A model may predict subsequent states from temporal correlations without identifying the effect of an intervention; providing an action input to the predictor does not by itself establish that actions are grounded in state changes [22], [79], [81]. A possible direction is to learn invariance selectively while covering a broader range of action magnitudes, durations, and targets during training. Causal representation learning offers relevant ideas, although their effectiveness in JEPA world models remains to be tested [180]. Evaluation should hold the observation history fixed, vary the candidate action, and check whether the predicted outcomes differ appropriately. Shuffled or substituted actions can further reveal whether the predictor relies mainly on scene correlations [181], [182]. Whether such sensitivity generalizes to actions outside the training policy remains open.

6.3 Long-Horizon Prediction and Uncertainty-Aware Dynamics

Accurate one-step prediction does not ensure reliable rollouts. Teacher-forced training repeatedly conditions on observed states, whereas a free-running model must use its own predictions as later inputs; small early errors can thus shift the input distribution at subsequent steps [4], [5], [183], [184]. Multi-horizon training, periodic access to observations, and feedback-based correction of latent states are possible directions. Their effectiveness must be tested across horizons rather than inferred from one-step error.

Even without accumulated error, the same history and action sequence may admit several plausible outcomes. In that setting, deterministic L_1 or L_2 regression can average incompatible futures into a state that is itself implausible [1], [185], [186]. Probabilistic transitions, multiple hypotheses, and ensembles offer alternatives to point prediction [187], [188]. Diffusion Forcing provides a related way to model sequences over varying horizons, but its benefit as a JEPA predictor requires separate evaluation [189].

Producing several possible futures is not the same as estimating uncertainty correctly. Evaluation should consider coverage of plausible outcomes, calibration, and how both change with prediction horizon, alongside prediction error [190], [191]. When predictions are used for planning, uncertainty should also help determine when to act, request another observation, or verify a forecast. Closed-loop tests can show whether new observations correct a latent state that has drifted during rollout [54], [183], [192]. To test the contribution of prediction to control, future work should also compare the same closed-loop system with the learned rollout enabled, removed, or replaced under a matched planning budget.

6.4 Structured States and Compositional Generalization

Explicit object and relation representations can support prediction of local changes, but a fixed object partition is difficult to maintain through occlusion, the arrival of new entities, or continued interactions. A model may retain the broad semantics of a scene while losing track of an object's identity, a changed contact, or a relation that should persist. Object-centric learning and sequential entity tracking seek to discover and maintain entities without assuming a fixed partition, while geometrically equivariant models can help preserve correspondence under spatial transformations [193], [194], [195].

Applying these ideas to JEPA would require a predictor to update identities, attributes, and relations under actions and missing observations, rather than merely extract objects from individual frames. Occlusion, unseen object combinations, and changes in scene composition provide direct tests of this ability. The unresolved question is how to learn a trackable latent structure without imposing a brittle scene decomposition in advance.

6.5 Efficient Latent Prediction and Selective Verification

Latent world models can predict task-relevant states without reconstructing every observable detail, offering a computational advantage for planning [3], [4], [5], [9]. Related approaches such as TD-MPC2 also show that planning need not decode an image at every step [196]. The difficulty is that latent predictions are hard to inspect directly: a state with small feature-prediction error may still omit a change that matters for collision or action choice. Frozen readouts can test whether selected properties remain recoverable, but cannot be expected to reveal every decision-relevant error [197]. Prediction accuracy and task success should therefore be considered alongside failure detectability and the cost of detecting a failure [183], [198].

One possible compromise is to verify only consequential prediction branches. A system might invoke decoding,

physical constraints, an external simulator, symbolic checks, or safety shielding when uncertainty or the potential cost of error is high [199]. The trigger matters as much as the check itself: missed errors undermine reliability, while frequent verification erases the efficiency gained from latent prediction. Comparisons under matched computational budgets should report error-detection coverage, decision quality, latency, and verification overhead to establish when selective verification is worthwhile.

7 Conclusion

This survey reviewed JEPA and its development toward world models. We introduced the basic principles of JEPA and organized existing methods by their prediction tasks into five families: foundational or static, temporal, action-conditioned, multimodal, and structured or scene-centric JEPA. We also examined their capabilities in representation, prediction, intervention, and control (R/P/I/C), summarized commonly used datasets and evaluation metrics, and compared the reported performance of representative JEPA methods and related baselines. We reviewed applications in visual understanding, robotics, autonomous driving, and other areas, and discussed open problems and future research directions in representation learning, action-conditioned prediction, and long-horizon planning.

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